PGPG: COVERING THE PARAMETER SPACE OF GUITAR PLAYING FOR IMPROVED TABLATURE TRANSCRIPTION

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ABSTRACT

Automatic guitar tablature transcription suffers from limited data. We propose a method that generates synthetic training data by randomly sampling guitar playing parameters from distributions, covering the space of possible guitar configurations rather than rendering musical patterns found in guitar tablatures of real music. Using banks of pre-recorded real guitar tones, we create diverse polyphonic data. Models trained with our data method achieve competitive or state-of-the-art performance on real-world evaluation. We also introduce novel fine-grained metrics that capture string-level and chord-level behavior, providing deeper insights into fret localization accuracy and chord structure consistency that standard classification metrics cannot reveal. Our approach demonstrates that systematically covering the parameter space of guitar playing, rather than replicating musical plausibility in training data, can train robust tablature transcription models.

1. INTRODUCTION

Recent datasets have broadened guitar coverage along individual axes, including full-length songs [1–6], timbral diversity [6–9], playing techniques [10–14], and recording modalities [3, 15, 16]. Yet tablature-annotated guitar audio remains scarce and fragmented: GAPS offers real full-length recordings (14 h), but only of classical guitar [5]; GOAT adds amplifier timbres (29.5 h), but no real audio effects [6]; DadaGP is large-scale (26,181 songs, ≈1,150 h of notated music), but has no audio [17]; and SynthTab synthesizes DadaGP tablatures (6,700 h), gaining scale but not realism [18]. Real recordings also require costly string-fret annotation, and song-derived data inherits the stylistic biases of its repertoire, which transposition only partially mitigates given the fretboard range. Thus, no single resource combines scale, realistic timbres, and systematic fretboard coverage. Evaluation is a second issue: despite steady model improvements, AGTT progress is still measured almost solely with Precision, Recall, and F_1, which reveal little about detailed model behavior.

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To address these limitations, we introduce a procedural method that tackles both data and evaluation gaps. Extending [19], which assembled single-note recordings following song tablatures, our approach generates tablature-annotated audio without any musical repertoire, systematically covering the string-fret configuration space. Labels are correct by construction and volume is not limited by annotated music; acoustic diversity is instead bounded by the variability of the single-note samples. We also propose fine-grained metrics that capture both the frequency and magnitude of errors across strings and frets.

Our main contributions are: (1) an open-source framework for song-independent AGTT data generation, (2) fine-grained evaluation metrics beyond classification accuracy, and (3) empirical evidence that models trained exclusively on ≈84 h of our procedural data achieve performance competitive with, or superior to, those trained on tablature-derived datasets. All code has been publicly released to support reproducibility. [1]

2. METHODOLOGY

2.1 Data Generation

We generate synthetic data from a configurable parameter space of guitar configurations using the synthesis algorithm described below (Fig. 1).

Synthesis algorithm. We sample parameters to generate three data families (Fig. 1); note, silence, and chord durations are drawn uniformly, $d \sim \mathcal{U}(d_{\min}, d_{\max})$ and all notes are scaled by a gain $\gamma \sim \mathcal{U}(\gamma_{\min}, \gamma_{\max})$.

Monophonic: at each step, $s \sim$ Bernoulli(p_{silence}) yields silence; otherwise a (string, fret) pair is drawn. Block polyphony: chords of size $n \sim \mathcal{U}\{n_{\min}, n_{\max}\}$ on distinct strings alternate with silences, with frets of one of three types: *position-feasible* ($n \in [2, 6]$, span ≤ 5, ≤ 3 fingers, no unisons or open strings above fret 5), *non-barre* ($n \in [2, 4]$, span ≤ 5, no unisons), or *unconstrained* ($n \in [2, 6]$). Temporalized polyphony: each chord is an arpeggio ($a = 1$) or a strum ($a = 0$), $a \sim$ Bernoulli(p_{arpeggio}); an arpeggio plays its notes in an order π drawn uniformly over permutations, with spacing $g \sim \mathcal{U}(g_{\min}, g_{\max})$; a strum spreads its note onsets uniformly over $[0, \delta]$, $\delta \sim \mathcal{U}(\delta_{\min}, \delta_{\max})$.

[1] https://procedural-guitar.github.io/

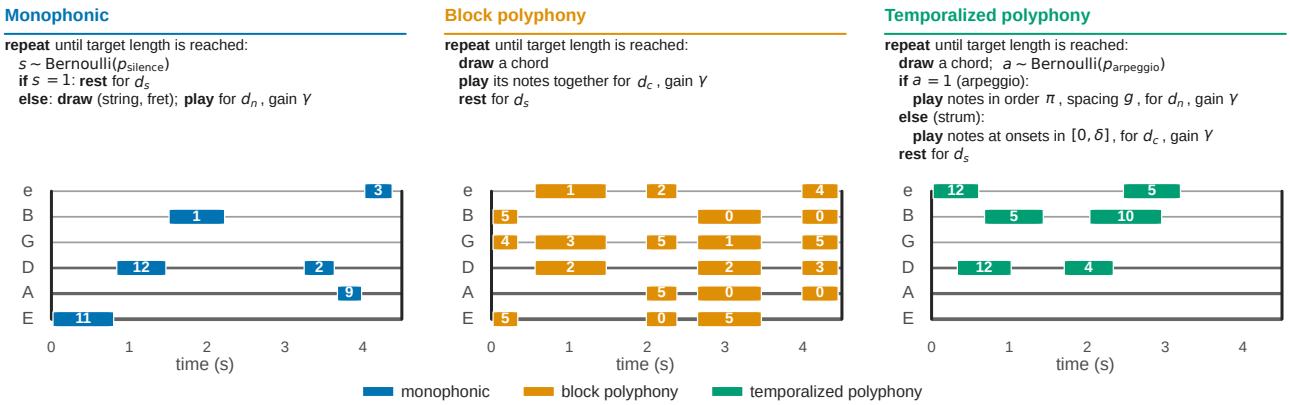


Figure 1. Data generation logic (top) and example tablature (bottom). Strings e–E run top to bottom, rectangle width is note duration, and the number inside is the fret. The three examples are independent excerpts, each with its own time axis.

2.2 Evaluation Metrics

Beyond standard global multi-pitch and tablature metrics [20], we introduce fine-grained metrics that measure proximity to ground truth at the string and chord levels.

Notation: Let $\mathcal{S} = \{1, \ldots, 6\}$ index strings and $t \in \mathcal{T} = \{1, \ldots, N\}$ index chord instances, with $\mathcal{T}_c = \{t : c_t = c\}$. For string s in instance t, y_s^t and $\hat{y}_s^t$ are the ground-truth and predicted states, encoded as $\varphi(\mathrm{U}) = 0$ (silent), $\varphi(\mathrm{O}) = 1$ (open), and $\varphi(\mathrm{F}_k) = k + 1$ (fret k). Let $A_t = \{s : \varphi(y_s^t) \neq 0\}$ be the active strings, $c_t = |A_t|$ the chord cardinality, and $f_s^t = \max(\varphi(y_s^t) - 1, 0)$ the fret number; $\hat{A}_t$, $\hat{c}_t$, $\hat{f}_s^t$ are defined analogously from $\hat{y}_s^t$. Instance-level metrics m_t are averaged over $\mathcal{T}$ (overall) and over $\mathcal{T}_c$, $c \in \{2, \ldots, 6\}$ (per cardinality).

String-level classification: Each string is classified at the *fret level* ($\mathcal{L} = \{\mathrm{U}, \mathrm{O}, \mathrm{F}_1, \ldots, \mathrm{F}_{19}\}$) and the *state level* ($\mathcal{L} = \{\mathrm{U}, \mathrm{O}, \mathrm{F}\}$, frets merged), with precision, recall, and F_1 computed per label.

Fret Error Distance (FED): FED measures *how far*, in frets, a prediction is from the truth. It is averaged over the strings active in the ground truth (open strings are fret 0):

$$\mathrm{FED}_t = \frac{1}{|A_t|} \sum_{s \in A_t} |\hat{f}_s^t - f_s^t|. \tag{1}$$

Fretboard Hamming Distance (FHD): the fraction of strings whose predicted label (silent, open, or specific fret) differs from the ground truth:

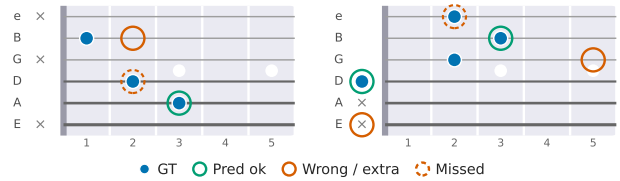
$$\mathrm{FHD}_t = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \mathbf{1}\big[\varphi(y_s^t) \neq \varphi(\hat{y}_s^t)\big]. \tag{2}$$

Chord Cardinality Accuracy (CCA): whether the number of sounding strings is correct, regardless of which strings or frets: $\mathrm{CCA}_t = \mathbf{1}[c_t = \hat{c}_t]$.

Chord Spacing Consistency (CSC): for $c_t \geq 2$, CSC compares the fret span $d_t^X = \max_{s \in A_t} f_s^t - \min_{s \in A_t} f_s^t$ and string span $d_t^Y = \max A_t - \min A_t$ with their predicted counterparts $\hat{d}_t^\alpha$ (computed on $\hat{A}_t$, $\hat{f}_s^t$):

$$\mathrm{CSC}_t^\alpha = \max\Big(0,\, 1 - \frac{|d_t^\alpha - \hat{d}_t^\alpha|}{K_\alpha}\Big), \quad \alpha \in \{X, Y\}, \tag{3}$$

where $K_X = F = 19$ (open = 0) and $K_Y = S - 1 = 5$ ($S = 6$ strings) are the largest attainable spans. We set $\mathrm{CSC}_t^\alpha = 0$ if $\hat{c}_t < 2$.



Metric	Calculation	Result
FED ↓	(0+2+1) / 3	1.00
FHD ↓	2/6 differ (D,B)	0.33
CCA ↑	GT = 3, Pred = 2	0
CSC-X ↑	1 − 1/19	0.95
CSC-Y ↑	1 − 0/5	1.00

Metric	Calculation	Result
FED ↓	(0+3+0+2) / 4	1.25
FHD ↓	3/6 differ (E,G,e)	0.50
CCA ↑	GT = 4, Pred = 4	1
CSC-X ↑	1 − 2/19	0.89
CSC-Y ↑	1 − 1/5	0.80

Figure 2. Per-frame metrics on two examples. Top: ground truth (blue dots) and predictions (rings; green: correct, orange: wrong/extra, dashed: missed). Bottom: calculation and values (arrows to better). Metrics: Fret Error Distance (FED), Fretboard Hamming Distance (FHD), Chord Cardinality Accuracy (CCA), and Chord Spacing Consistency for frets (CSC-X) and strings (CSC-Y).

The metrics are complementary: FED captures how far wrong frets are, FHD the fraction of wrongly predicted strings, CCA the sounding-string count, and CSC the preservation of chord span. Fig. 2 shows example frames; reported values average over all evaluated frames.

3. EXPERIMENTS AND RESULTS

3.1 Experimental Setup

Models and Datasets. We compare three models sharing the same backbone architecture: (i) TabCNN [20], which we reimplemented and trained from scratch on GuitarSet [1] following the original protocol; (ii) Pedroza (2024) [19] tablature trained model, for which we use the best publicly released pretrained checkpoint *as is*, without retraining or fine-tuning; and (iii) our proposed PGPG (Procedurally Generated Polyphonic Guitar), trained by us. All variants are validated on GuitarSet [1] (real acoustic guitar recordings) and evaluated

Table 2. Comparison of models using the global metrics introduced in TabCNN [20]. TabCNN was reimplemented from scratch following the original architecture and training protocol, while Pedroza (2024) [19] was evaluated using its publicly released pretrained checkpoint, without retraining. The left and middle column groups report multi-pitch and tablature metrics, respectively; the rightmost column reports the Tablature Disambiguation Rate (TDR), the ratio of correctly predicted tablature notes to correctly predicted pitches.

Model	**Multi-Pitch**			**Tablature**			
	Precision (↑)	**Recall** (↑)	**F_1 Score** (↑)	**Precision** (↑)	**Recall** (↑)	**F_1 Score** (↑)	**TDR** (↑)
PGPG (proposed)	**0.906**	**0.750**	**0.821**	0.645	**0.562**	**0.601**	0.712
Pedroza 2024 [19]	0.904	0.711	0.788	0.644	0.528	0.576	0.732
TabCNN [20]	0.883	0.496	0.635	**0.655**	0.369	0.472	**0.741**

Table 3. Per-string classification at the *fret level* (silent, open, or a specific fret) and *state level* (silent, open, or fretted). Precision, recall, and F_1 are support-weighted within each string, then averaged over the six strings. TabCNN was reimplemented from scratch; Pedroza (2024) [19] uses its public pretrained checkpoint without retraining.

Model	**Fret level predictions**			**State level predictions**		
	Precision (↑)	**Recall** (↑)	**F_1 Score** (↑)	**Precision** (↑)	**Recall** (↑)	**F_1 Score** (↑)
PGPG (proposed)	**0.817**	0.798	**0.794**	**0.835**	0.829	**0.826**
Pedroza 2024 [19]	0.806	**0.811**	0.792	0.826	**0.832**	0.825
TabCNN [20]	0.743	0.766	0.727	0.787	0.787	0.761

out-of-domain on EGSet12 [19] (twelve solo electric guitar tracks, avg. duration 31.65 s) spanning diverse genres, tonalities, hardware, and techniques.

Model and Training. TabCNN maps spectral inputs to six 21-class string predictions (unused, open, frets 1–19). Following [20], we train for 8 epochs using AdaDelta (lr= 1, batch size 128, $\rho = 0.9$, $\varepsilon = 10^{-6}$ selected empirically), selecting the model with minimal validation loss. We also train ablated variants on individual data subsets to quantify each category's contribution.

Data Setup. Audio is procedurally generated using empirical parameter distributions (Table 1) and single-note samples from IDMT-SMT [21], EGFXSet [7], and GuitarTECHs [3], maintaining timbral realism while freely covering the string-fret space. The dataset totals 674 files (≈84 h): 514 non-barre (≈52 h), 100 monophonic (≈15 h), and 60 polyphonic (≈17 h). Unconstrained sampling yielded poor results, so our final dataset relies primarily on position-feasible and non-barre chords (Section 2).

Table 1. PGPG synthesis parameters for procedural data generation: silence probability and duration, note and chord durations, arpeggio probability, note order and spacing, strum offset, and per-note gain.

Symbol	**Value / Range**	**Description**
p_{silence}	0.2	Silence probability
d_s	$\mathcal{U}(0.1, 5.0)$	Silence duration
d_n	$\mathcal{U}(0.1, 1.0)$	Note duration
d_c	$\mathcal{U}(0.23, 1.2)$	Chord duration
p_{arpeggio}	0.7	Arpeggio probability
π	$\mathcal{U}(S_n)$	Arpeggio note order
g	$\mathcal{U}(0.05, 0.2)$	Arpeggio note spacing (s)
δ	$\mathcal{U}(0.05, 1.0)$	Strum offset (s)
γ	$\mathcal{U}(0.6, 0.9)$	Per-note gain scaling

Table 4. Comparison of models using the proposed fine-grained metrics: Fret Error Distance (FED), Fretboard Hamming Distance (FHD), Chord Cardinality Accuracy (CCA), and Chord Spacing Consistency (CSC) along the fret (X) and string (Y) axes. Values are overall means.

Model	**FED** (↓)	**FHD** (↓)	**CCA** (↑)	**CSC (X,Y)** (↑)
PGPG (proposed)	**2.469**	0.343	0.596	0.899, **0.868**
Pedroza 2024 [19]	2.680	**0.326**	**0.602**	**0.930**, 0.865
TabCNN [20]	3.374	0.426	0.442	0.910, 0.802

3.2 Transcription Performance

Standard Metric Analysis. PGPG outperforms prior work on most metrics (Table 2), leading in multi-pitch F_1 (0.821 vs. Pedroza 2024: 0.788, TabCNN: 0.635), driven by higher recall (0.750 vs. 0.711, 0.496), and in tablature F_1 (0.601 vs. 0.576, 0.472). Only in tablature precision and TDR does TabCNN rank first (0.655 and 0.741 vs. 0.645 and 0.712 for PGPG).

Fret- and String-Level Analysis. PGPG and Pedroza 2024 are nearly tied at fret level (F_1 0.794 vs. 0.792) and state level (0.826 vs. 0.825), both clearly above TabCNN (Table 3). PGPG has higher precision (0.817, 0.835), while Pedroza 2024 slightly edges it in recall (0.811 vs. 0.798; 0.832 vs. 0.829).

Fine-Grained Metric Analysis. Fine-grained metrics (Table 4, Figure 3) reveal nuances that standard evaluation misses. PGPG achieves the best Fret Error Distance (2.47), with the largest gains on strings 1, 5 and 6, and matches Pedroza 2024 in string-axis Chord Spacing Consistency (0.87). It trails Pedroza 2024 slightly in Fretboard Hamming Distance (0.34 vs. 0.33), Chord Cardinality Accuracy (0.596 vs. 0.602) and fret-axis spacing (0.90 vs. 0.93). However, PGPG's spacing consistency is more sta-

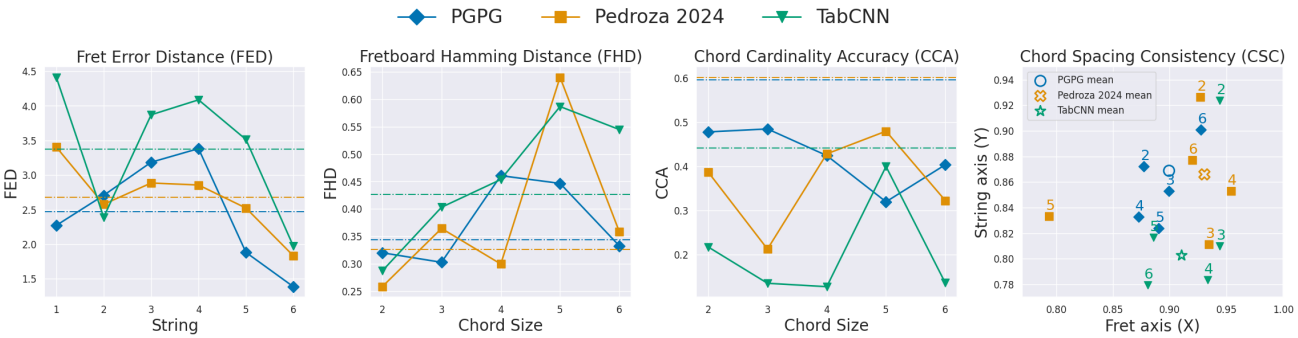


Figure 3. Fine-grained metric breakdown (left to right): FED per string, FHD per chord size, CCA per chord size, and CSC (string- vs. fret-axis span) per chord size, , where the number next to each CSC point denotes the chord size (number of active strings). Models are distinguished by color and marker. Dash-dotted lines and outlined markers indicate the global means pooled over all chords (not the mean of the per-group values); these values are also reported in Table 4.

Table 5. Ablation study of PGPG training data variants: MS (Monophonic Sequences), NBC (Non-Barre Chords), and TPPFC (Temporalized Polyphony Position-Feasible Chords). Metrics include pitch and tablature F_1, TDR, per-string F_1 (fret/state), and FED, FHD, CCA, and CSC.

Dataset	Pitch F_1 (↑)	Tab F_1 (↑)	TDR (↑)	Fret F_1 (↑)	State F_1 (↑)	FED (↓)	FHD (↓)	CCA (↑)	CSC(X,Y) (↑)
MS	0.373	0.268	**0.725**	0.635	0.645	3.736	0.614	0.475	0.367, 0.393
NBC	**0.754**	**0.488**	0.634	**0.728**	**0.761**	**2.247**	**0.429**	**0.732**	**0.917**, **0.816**
TPPFC	0.485	0.309	0.620	0.667	0.694	4.152	0.590	0.716	0.710, 0.664

ble across chord sizes (2–6 notes) on both axes, with its points clustered in a narrow band, whereas the other models vary considerably with chord size.

3.3 Ablation Study

Table 5 shows distinct contributions across training components. Non-Barre Chords (NBC) is the strongest single subset, leading in pitch F_1 (0.754), tablature F_1 (0.488), and FED (2.247), underscoring its role in chord recognition and fret localization. Conversely, Monophonic Sequences (MS) underperform overall but achieve the highest TDR (0.725), aiding note disambiguation. Temporalized Polyphony Position-Feasible Chords (TPPFC) offer moderate, supplementary signals. Thus, chord-based data drives core transcription performance, while monophonic data refines string-fret precision.

4. DISCUSSION

PGPG, trained only on procedurally generated data, outperforms TabCNN [20] and is competitive with Pedroza 2024 [19], which is trained on tablature-derived data. Its main advantages are in recall and fret localization, suggesting that controlled string-fret sampling helps recover and place more notes correctly. Its small gaps in mean chord-level metrics are partly offset by stable chord-structure behavior across polyphony levels, whereas the other models' consistency depends strongly on chord size.

PGPG's lower TDR (0.712 vs. 0.732 and 0.741) is consistent with it recovering more correct pitches, which enlarges the TDR denominator and exposes more fingering ambiguities. Conversely, TabCNN's high TDR coincides with its poor multi-pitch recall (0.496), as TDR inflates when few pitches are predicted. TDR should therefore be read alongside recall.

Ablations show complementary data strengths: non-barre polyphony drives core accuracy, temporalized polyphony provides moderate support, and monophonic data preserves TDR and disambiguation. Combining them in a single procedural dataset enables PGPG to rival tablature-trained models, supporting purely synthetic training for AGTT.

5. CONCLUSION

A novel approach is presented for data generation aimed at training AGTT models. Unlike previous work, it does not rely on song tablatures, but instead uses procedural elements that systematically cover the guitar's parameter space, including configurations beyond typical human playability. This makes the method scalable and free of the genre and playing-style biases inherent to tablature repositories. Data comprises monophonic sequences, block polyphony, and temporalized polyphony.

We also introduce fine-grained AGTT metrics that track error magnitude and prediction consistency at the fret, string, and chord levels, surfacing model distinctions missed by standard classification metrics.

We expect these results to deepen the understanding of guitar transcription's complexities, and to establish parametric coverage as a serious alternative to tablature-based data. Future work will combine parametric and tablature datasets and extend to new modalities.

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